A Project Report

On

Business Function Agentic Workflows: HR, Marketing, Podcast Research & Enterprise Integrations

Submitted in partial fulfilment of the requirement of

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For the Degree of

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in

Computer Engineering

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CERTIFICATE

This is to certify that the project entitled

Business Function Agentic Workflows: HR, Marketing, Podcast Research & Enterprise Integrations

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Declaration

We declare that this written submission for BE Major Project Declaration entitled "**Business Function Agentic Workflows: HR, Marketing, Podcast Research & Enterprise Integrations**" represent our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declared that we have adhere to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause for disciplinary action by institute and also evoke penal action from the sources which have thus not been properly cited or from whom paper permission have not been taken when needed.

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Abstract

Business teams in HR, marketing and podcast production spend much of their time on repetitive work such as screening resumes, rewriting posts for each channel and researching guests. Existing tools mostly store records or match keywords, and their AI features give opaque results that are difficult to explain or audit. Our project aims to build an integrated agentic AI platform in which a workflow router coordinates specialised LLM agents for recruitment, marketing, podcast guest research and enterprise integrations. It combines retrieval-augmented generation with rubric-based, evidence-backed scoring and secure connectors to Zoho, Workday and Greenhouse-style systems. The system generates rubrics from job descriptions, screens and interviews candidates, drafts channel-specific content and prepares source-backed guest briefings. By placing human approvals, PII protection and audit logs around consequential actions, the platform helps organisations automate routine work while keeping people in control of final decisions.

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Chapter 1

Introduction

1.1 Background

Even in the most modern and efficient business teams today, there is still a considerable amount of repetitive work that can't be entirely delegated. A recruiting HR manager may review hundreds of resumes for one job and find that if they are doing the same thing each time; it takes hours to do the job. A marketing team writes the same campaign slogan 5 times, each for different channels: LinkedIn, Instagram, Newsletter, etc. A podcast producer puts together an interview with a guest over the course of a few hours by collating interviews, articles and social profiles spread across the web. None of this is very creative, but it is the time people waste on the decisions that really count.

The software these teams were using didn't really help as much. ATS software, HR software or CRM software is great at storing and retrieval of records, not so good at handling unstructured data like a resume, a job description in natural language or a long article. They have rigid rules and will not work with inputs that do not fit its requirements. In recent years, the large language model has evolved, making it possible to achieve much more here: [1, 2, 3]. Such models can understand unstructured text, think about it, invoke external tools and share its output with other agents in a pipeline. Our project, in collaboration with InfinityPool (Shankh), takes this ability to the business processes where the software doesn't match the real requirement of the teams.

1.2 Motivation

The current challenges with tools are no mere nuisance. When it comes to hiring, keyword screening has been found to be useful in rewarding those candidates who know how to craft a resume rather than the best candidates for the position Gan et al. [4] showed that an LLM-based pipeline can

screen resumes eleven times faster than a human and reach a respectable F1 score, but their system was tested only on IT-sector data and could not generate rubrics from a job description or conduct an interview. Wilson and Caliskan [5] found that embedding-based screening favoured White-associated names in 85 out of every 100 cases and that Black male candidates were disadvantaged in up to 100% of test cases. Mu et al. [6] then showed that adversarial content hidden inside a resume could manipulate LLM screening more than 80% of the time, and that the best available defences reduced this by only around a quarter. If a recruiter is given a ranked shortlist without a note, there is nothing they can do to detect these mistakes until they impact a real life employee.

In marketing, content that is sent out without a proper review can cause harm to a brand that can take months to fix. Research for podcasts can also result in a reputation risk if the host or podcast relies on unsourced or misattributed claims [10, 8]. Whether it's the social, or the finance, or the other aspect of the business, the data required to work is in different systems with different logins, API, rate limits, quirks etc. The task of compiling everything in one place is quite a full time occupation.

What teams really need is something that can process the repetitive parts end-to-end, something which could be explained unambiguously to a person for purposes of review and trusted, and then wait for the person to sign off on it before some important thing occurs. This project is meant to fill that void.

1.3 Aim and Objectives

Aim :

Business Function Agentic Workflows's mission is to create an integrated multi-agent AI platform that handles repetitive, high-volume components of HR recruitment, marketing content production, podcast guest research and enterprise data integration — leaving a human reviewer to handle each decision that matters. The platform is built with a series of specialized LLM agents, approved by the approval gates, and supported by a retrieval-augmented knowledge layer, with audit logs and controls in between.

Objectives :

The specific objectives of the project are:

- To build HR workflows that generate weighted rubrics from a job

description, screen and score resumes with cited evidence, conduct AI-assisted interviews and produce a ranked shortlist for the recruiter to review, alongside assistants that answer onboarding, leave and HR policy questions.

- To build marketing workflows that turn a theme or source content into a content calendar, channel-specific drafts and audience segments, with every piece of content held for human approval before it goes anywhere.
- To build podcast research workflows that gather public sources about a guest, index them for retrieval, and produce a structured briefing in which every claim is linked to a source and labelled as a verified fact, an allegation or an opinion.
- To build a connector layer that communicates with Zoho, Workday-style and Greenhouse-style systems through authenticated, rate-limit-aware, idempotent API calls, with human approval required before any write operation goes through.
- To enforce consistent governance across all workflows through a compliance reviewer that checks for PII, prompt injection and unsupported claims, an approval queue that holds every consequential output, and an audit log that records every prompt, tool call, source and reviewer action.
- To package the entire platform as a reusable workflow framework and measure its performance on accuracy, hallucination rate, latency, cost and task completion.

Chapter 2

Study Of the System

2.1 Literature Survey

Building an agentic AI platform for business functions means drawing on work from several overlapping areas of research. We looked at how multi-agent frameworks are structured, how AI has been applied to hiring and whether it can be trusted to do so fairly, how marketing content generation and hallucination reduction have been approached, and what the state of the art says about tool use, privacy and governance. The picture that emerges is one of genuine progress alongside well-documented gaps, and those gaps are what this project tries to close.

Multi-Agent Orchestration. In AutoGen, Wu et al. [1] proposed a framework that enables users to write applications using multiple agents, with each agent using an LLM, a human, or an external tool. Guo et al. [2] followed with a very general overview of multi-agent systems based on LLM, including the profile of agents, agent communication, and the development of agent capabilities over time. Yao et al.[3] found that the alternating strategy of reasoning steps and tool actions yields better results than either. None of these can provide an out-of-the-box approach to a domain-specific business workflow. They provide us with building blocks, but it's going to take some work to design the plumbing for rubric-based scoring, as well as human approval gates and decision-level audit logs. **AI-Powered Recruitment and Bias.** The recruitment literature

makes a sobering case for caution. Gan et al. [4] showed that an LLM-based pipeline can screen resumes eleven times faster than a human and reach a respectable F1 score, but their system was tested only on IT-sector data and could not generate rubrics from a job description or conduct an interview. Wilson and Caliskan [5] found that embedding-based screening favoured White-associated names in 85 out of every 100 cases and that Black male candidates were disadvantaged in up to 100% of test cases. Mu

et al. [6] then showed that adversarial content hidden inside a resume could manipulate LLM screening more than 80% of the time, and that the best available defences reduced this by only around a quarter. Stuart et al. [7] took things further by building an AI interviewer that updates its belief about a candidate's competencies after each answer, recovering accurate ability profiles with 76.1% accuracy. Together these papers describe a picture where AI recruitment tools can save time but cannot yet be trusted without human oversight, bias monitoring and adversarial scanning at every step.

Content Generation and Marketing. Liu et al. [8] demonstrated that retrieval-augmented generation can produce keyword-specific ad copy that outperforms template-based content in online A/B tests, with an automated evaluation approach that agrees with human reviewers nearly 90% of the time. The limitation is scope: the system was designed for search advertising rather than for brand-consistent posts across multiple social channels, and it does not include an approval step before content goes live. Our marketing workflow extends the RAG-based generation idea to channel-specific branded content and adds an explicit queue where a human must approve every draft before it is published.

Retrieval-Augmented Generation and Hallucination Reduction. To evaluate whether to retrieve or not, and to judge whether the retrieved facts are accurate or not, Asai et al. [9] trained a model to do both using special reflection tokens, achieving better factuality scores than baseline retrieval-augmented approaches. In another approach, Dhuliawala et al. [10] drew the model to draft an answer, created verification questions and then refined the answer according to the verification results, resulting in fewer hallucinations on some benchmarks. Both of these ideas are actually flawed in practice; either they require custom training for the machine to learn the new information or they only work if the information is something the machine already knows. Our podcast research agent takes the retrieve-then-verify concept, and implements it with prompting: Any statements that are made are made from a public source, and marked as a "verified fact", an "allegation" or an "opinion" before being directed to the host.
Tool Use and Enterprise API Integration. Schick et al. [11] showed that a language model can learn which API to call and when, purely from a small number of demonstrations, without supervised training. Patil et al. [12] went further, fine-tuning a model to write accurate API calls even when the documentation changes after training. Both results are impressive, but

neither system can chain multiple authenticated calls, handle rate limits or decide whether a particular write operation should require human approval before it goes through. Enterprise integrations with Zoho, Workday and Greenhouse require all of these things, which is why our connector layer adds OAuth scopes, retries, idempotency keys and a mandatory approval step for any call that writes data.

Privacy and Governance. Lukas et al. [13] conducted an empirical study of the extent to which a language model can expose PII sequences and discovered that even with techniques such as differential privacy, approximately 3% of the PII sequence is still exposed. Mökander et al. [14] came up with a governance blueprint for LLMs at organisational, model and application levels but did not include how to do application-level auditing in software in the paper. We take a proactive approach to PII risk by masking all PII data in a candidate's response before it is sent to an LLM and limiting the data that is logged. In this case, application audit layer is not a design concept but a real implementation: every prompt, tool call, source, draft version, reviewer decision and final output are recorded and tracked.

Summary of Related Research Papers

S.no	Paper	Main Idea / Objective	Tools / Tech Used	Gaps Identified
1	AutoGen (2024)	Framework for building applications from multiple conversable agents that combine LLMs, humans and tools.	Multi-agent conversation framework; human-in-the-loop support.	General-purpose; no HR or marketing workflow, rubric scoring, approval gates or audit design.
2	LLM Based Multi-Agents: A Survey (2024)	Reviews domains, agent profiling, communication, capability growth and benchmarks of LLM multi-agent systems.	Systematic literature review; open-source tracking repository.	Survey only; no concrete business-function system, governance or integration.
3	ReAct (2023)	Interleaves reasoning traces with tool actions to reduce hallucination and improve decision-making.	Few-shot prompting; Wikipedia API; HotpotQA, FEVER, ALFWorld, WebShop.	Single agent with read-only tools; no approval or audit for actions with real effects.
4	LLM Agents in Recruitment (2024)	Agent pipeline that classifies, grades, summarises and selects resumes.	Fine-tuned LLaMA2; GPT-3.5 and GPT-4 baselines; BERT, T5.	IT resumes only; labels mostly from GPT-4; no JD-to-rubric generation or interview stage.
5	Bias in Resume Screening via LM Retrieval (2024)	Audits embedding-based screening for gender, race and intersectional bias.	Massive Text Embedding models; retrieval-based audit on 500+ resumes.	Measures bias; proposes no bias-aware screening or human review step.
6	Adversarial Vulnerabilities in Resume Screening (2025)	Shows hidden instructions and invisible content can manipulate LLM screening.	12 LLM variants; prompt defence; FIDS (LoRA).	Defences cut attacks by only 10–26%; no pipeline-level scanning or human checkpoint.
7	Beyond the Resumé (2026)	Rubric-aware LLM interviewer that updates beliefs about competencies after each answer.	GPT-5; belief updating; metamorphic testing; Chainlit.	Simulated interviews only; three-level rubrics; no rubric generation, ATS link or HR approval.
8	LLMs for Marketing Content Generation (2025)	Generates keyword-specific ad copy and evaluates it at scale.	RAG; LLM-as-a-Judge; rule-based metrics; A/B testing.	Ad copy only; no multi-channel brand posts or approval workflow.
9	Self-RAG (2024)	Model learns when to retrieve and critiques its own output with reflection tokens.	Fine-tuned Llama2 7B/13B; retriever; reflection tokens.	Needs custom training; not evaluated on high-risk research about real people.

S.no	Paper	Main Idea / Objective	Tools / Tech Used	Gaps Identified
10	Chain-of-Verification (2024)	Drafts, plans and answers verification questions, then revises to reduce hallucination.	Multi-stage prompting; factored verification.	Uses only the model's own knowledge; no external sources or claim labels.
11	Toolformer (2023)	LLM teaches itself when and how to call external APIs.	GPT-J 6.7B; self-supervised API-call annotation.	One call per example; no tool chaining, authentication or write actions.
12	Gorilla (2024)	Fine-tuned LLM writes accurate API calls with retrieval over documentation.	LLaMA-7B; BM25 and GPT-Index retrievers; APIBench; AST matching.	Public ML APIs only; ignores permissions, PII and approval of write operations.
13	PII Leakage in Language Models (2023)	Defines and measures PII extraction, reconstruction and inference attacks.	GPT-2 fine-tuning; black-box attacks; differential privacy.	Covers training-data leakage; not PII in run-time prompts, logs and tools.
14	Auditing LLMs: A Three-Layered Approach (2024)	Blueprint of governance, model and application audits for LLMs.	Conceptual governance framework.	Conceptual; no implementation of application-level audit trails.

2.2 Existing Systems

There are a number of commercial tools currently addressing the overlapping portions of the problem, but nothing available that could integrate HR screening with marketing content creation, podcast research verification and enterprise automation in one managed workflow. While all have their strengths in recruitment (HireVue, Greenhouse, and Zoho Recruit), marketing (Jasper), general research (Perplexity Deep Research), and automation (n8n), they also have weaknesses in other critical functions including rubric generation, claims supported by evidence, source accountability, auditability, approval gates, and end-to-end business workflow orchestration. To put it simply, the existing market is filled with point solutions for domains, but not a single agentic platform for business operations that is policy-aware.

Chapter 3

Proposed System

3.1 Problem Statement

Today's tools were designed for an era when information was stored in structured databases and workflows were governed by hard-coded rules. Posting a job meant setting up a job and filling out a spreadsheet of applications, then individually reading through them. Marketing involved developing a template and filling the same in for every channel. Research involved having a number of browser windows and making notes. As the amount of work has increased significantly, and the risks of failure have become more obvious, these processes have not significantly changed.

A recruiter using a keyword filter who looks at 1,000 resumes will overlook good candidates that did not state their experience in the proper manner. A screening system without scores to explain them means that the recruiter will not be able to detect errors or bias. Without a review process, a marketing team can find itself on the verge of a brand crisis after publishing just one bad post. If a podcast host puts together a guest briefing without verifying information from unreliable sources, then he could be spreading false information or defamation on the air. In each instance, the information the team is after is stored in a different type of application or system than the one they're searching, each with its own login, and each with its own search methods. What is missing is a system that takes over the repetitive, high-volume parts of these workflows, explains every output clearly enough that a person can check it, and then waits for that person's approval before doing anything that cannot be undone.

3.2 Scope

This project covers four business workstreams, defined in collaboration with our industry partner InfinityPool (Shankh): HR recruitment and HR assistance, marketing content operations, podcast guest background re-

search, and enterprise integrations with Zoho, Workday and Greenhouse-style platforms. In all four, a common workflow router, a shared retrieval-augmented knowledge layer, approval gates at every "consequential" step to be filled by humans, and a complete audit log, with a privacy protection of PII, all of this as a reusable framework.

Clear demarcation of what the system does and doesn't do. It helps people, it doesn't make hiring decisions, rejection decisions, salary decisions or disciplinary decisions on their behalf. It stores content that is for approval and it publishes nothing that hasn't been approved. All development is done using synthetic data, public data or sanitised data, and not live production credentials, or sandbox APIs. It is planned to be expanded to other business functions later, for example finance and sales; however, this is not covered by the current scope.

3.3 Proposed System

Business Function Agentic Workflows is a modular multi-agent platform built around a single idea: business workflows should be fast and consistent where the work is repetitive, and human where the decisions matter.

System Overview

All interactions begin at a Web dashboard. A request is triggered by a user or by an incoming business events and a workflow router categorizes it and routes it to the appropriate set of agents. Those agents retrieve information from a shared context store and invoke enterprise connectors as necessary. Any action with a consequence – sending an email to a candidate, publishing a post, writing a record back to an HR system – is put into an approval queue, and is only executed once it is approved. It is examined by a human being, approved and/or edited by a human, and only then does the system act. All exchanges are logged in an audit log.

Key Components

- **Web Dashboard** – A Next.js interface where HR managers, marketing teams and researchers interact with the system: creating jobs, reviewing candidate rankings, editing content drafts and clearing the approval queue. Candidates have a separate public-facing apply page.
- **Workflow Router** – Classifies each incoming request as an HR, marketing, podcast research or integration task and hands it to the match-

ing set of agents.

- **HR Agents** – A set of specialised agents covering rubric generation from job descriptions, resume parsing and adversarial scanning, claim verification, evidence-backed screening, interview question generation, live AI interviewing, candidate ranking and follow-up. Separate assistants handle employee queries about onboarding, leave, attendance and HR policy.
- **Marketing Agents** – Agents that turn a theme or source content into a content calendar, campaign and creative briefs, audience segments and channel-specific social copy. All drafts wait in an approval queue before going anywhere.
- **Podcast Research Agents** – Agents that gather public sources about a guest, index them for retrieval, build a structured briefing and label every claim as a verified fact, an allegation or an opinion. Sensitive-topic guardrails are applied before the briefing reaches the host.
- **Integration Agent and Connector Layer** – A unified connector interface for Zoho People, Zoho CRM and Recruit, Workday-style HRIS, Greenhouse-style ATS, CRM and email, calendar and social media APIs. Handles OAuth, rate limits, pagination, webhooks, retries and idempotency so that agents do not need to know the details of each system.
- **Knowledge (RAG) Layer** – Vector store with company policies, brand guidelines, job descriptions, approved rubrics, past campaigns, guest research sources. Agents look up in this store, they never use 'memory' alone.
- **Compliance Reviewer and Guardrails** – Acts as the bridge between the output from each agent and the approval queue. Looks for PII that shouldn't be part of a prompt or log, for prompt injection in user-contributed content, and for unfounded claims with a cited source. Raises these issues for human correction rather than letting them go unnoticed.
- **Approval Layer** – It holds the rubrics, hiring decisions and drafts of contents until a human acts on them. Nothing goes ahead of this step without approval.

- **Audit Layer and Data Store** – Supabase storage for every job, candidate, score, draft and report, including a versioned log of every prompt, tool call, source, draft, reviewer and final output. Secured by role-based access controls.

Workflow of the Proposed System

Upon receipt of the request, the workflow router categorize it and route it to the right specialists. The specialists get the required information from the knowledge base and call the enterprise connector where required. All outputs that have an implication or impact are not immediately picked up but stored in the approval queue. The compliance reviewer examines it for PII leaking, prompt injection and unsupported claims before it is passed to the reviewer. The reviewer human analyzes it, approves, edits or declines it. Approved actions are sent through the connector layer. The audit log records every step from initial prompt to final output.

Chapter 4

Design Of the System

4.1 System architecture

4.1.1 Data Flow Diagrams

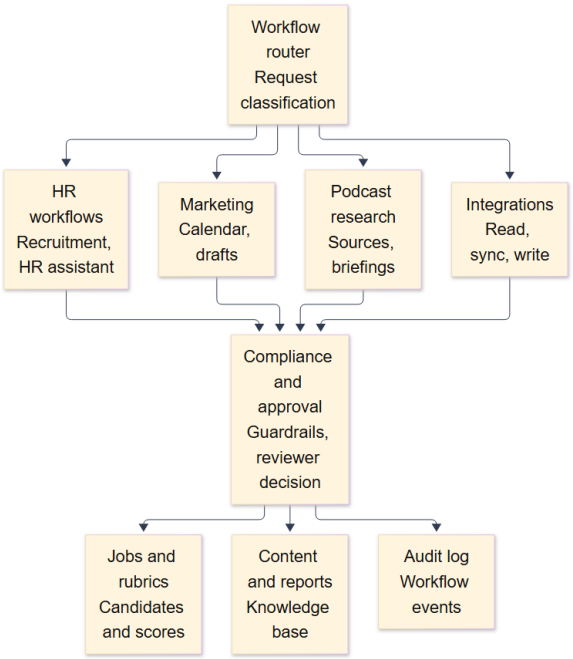


Figure 4.1.1 DFD Level 0 of the System

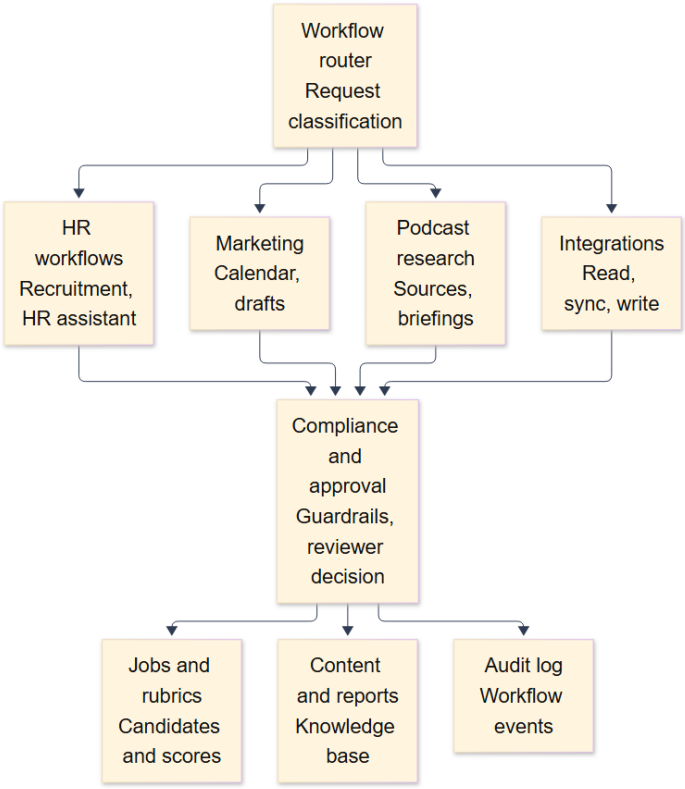


Figure 4.1.2 DFD Level 1 of the System

4.1.2 UI/UX Diagram

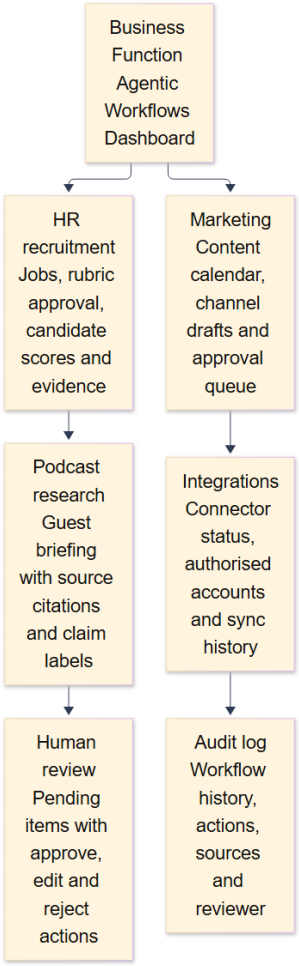


Figure 4.2.1 UI/UX Diagram of the System

4.1.3 Block Diagram

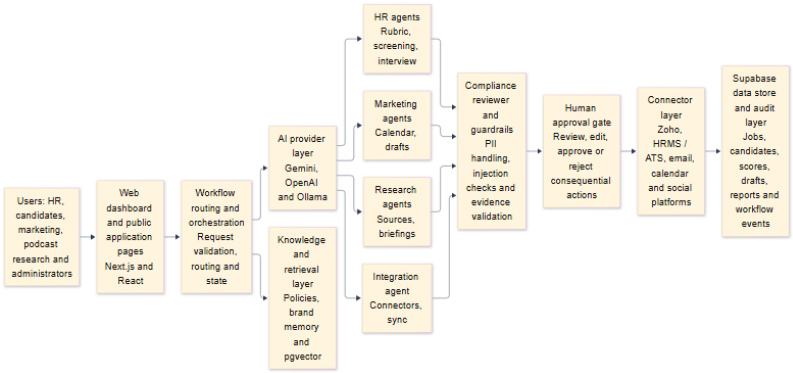


Figure 4.2.2 Block Diagram of the System

4.1.4 Agent Workflow Diagrams



Figure 4.2.3.1 HR Agent Workflow of the System



Figure 4.2.3.2 Marketing Agent Workflow of the System

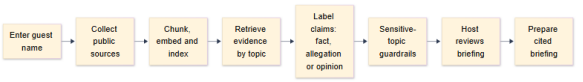


Figure 4.2.3.3 Podcast Research Workflow of the System

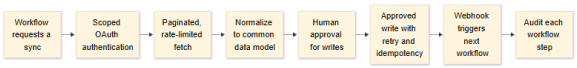


Figure 4.2.3.4 Enterprise Integration Workflow of the System

Chapter 5

Result and Discussion

5.1 Screenshots of the System

5.1.1 HR Job Creation and Rubric Approval

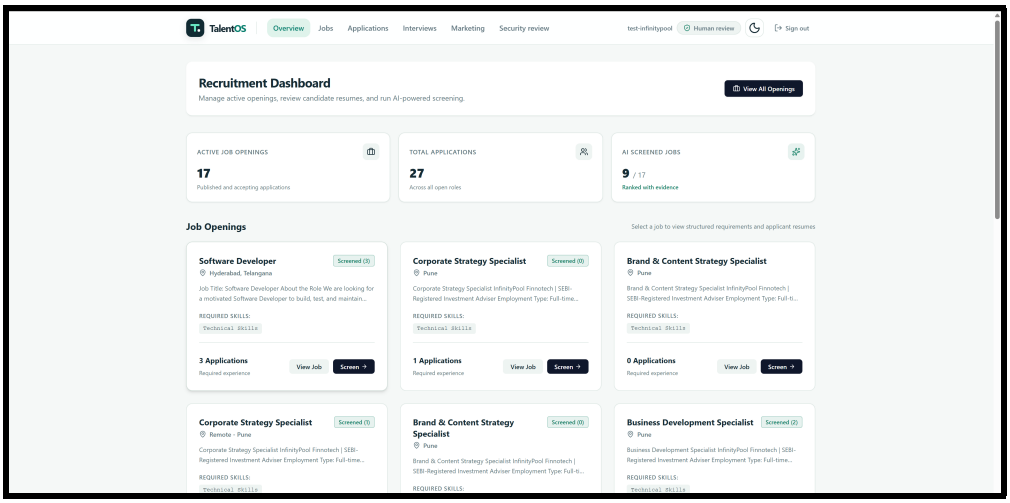


Figure 5.1.1 HR Job Creation and Rubric Approval of the System

5.1.2 Candidate Screening Results

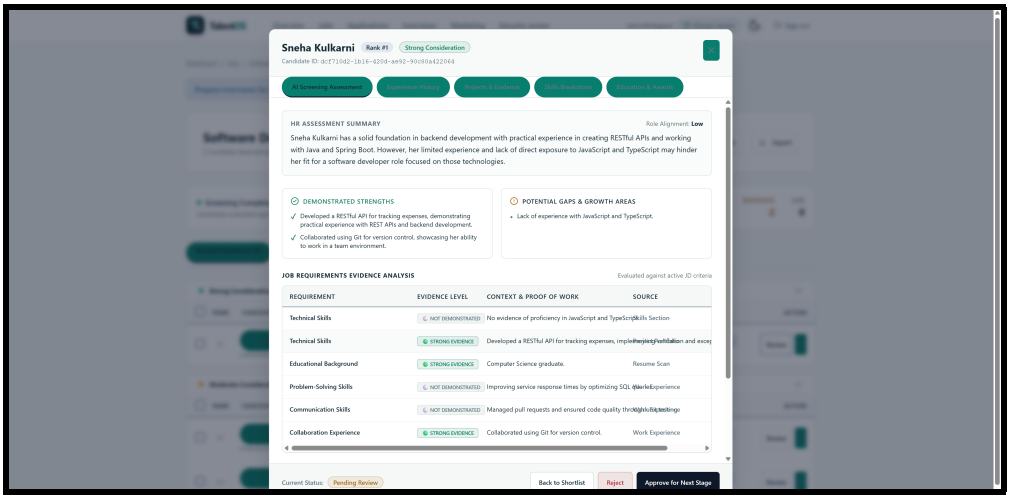


Figure 5.1.2 Candidate Screening Results of the System

5.1.3 AI Interview and Final Ranking

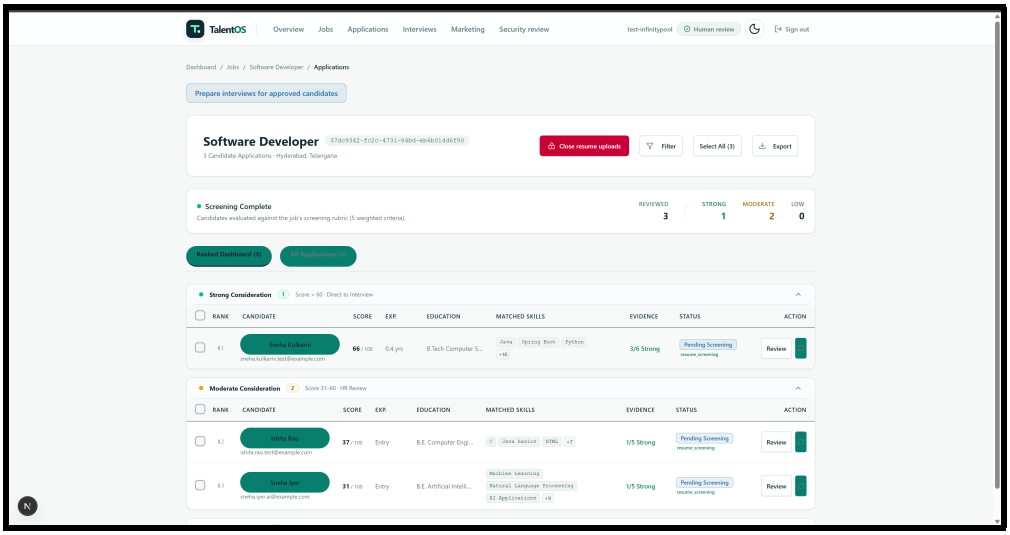


Figure 5.1.3 AI Interview and Final Ranking of the System

5.1.4 Marketing Calendar and Approval Queue

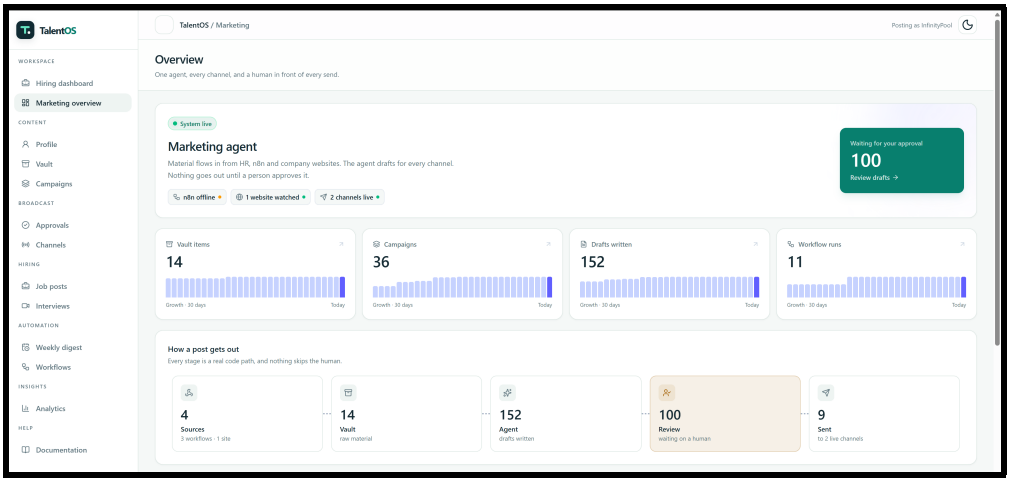


Figure 5.1.4 Marketing Calendar and Approval Queue of the System

Chapter 6

Conclusion & Future Scope

Conclusion

Business Function Agentic Workflows unites the world of record keeping business tools and fully autonomous AI by integrating specialised LLM agents with retrieval-grounded knowledge, enterprise connectors and human approval at each critical stage. Our project automates the drudgery of tasks like HR hiring, marketing content and podcast guest research, as well as enterprise integrations, while ensuring that every output is explainable through rubric evidence and citations. Through approval gates, PII safeguarding and complete audit trails in a reusable workflow framework, our solution allows organisations to work faster whilst ensuring humans retain responsibility for the end decision.

Future Scope

Other possible add-ons include: additional business areas (finance, sales & customer support), voice & multilingual interview support, screening outcome fairness auditing, more enterprise connectors, continuous evaluation dashboards and more robust enterprise access-control & retention policies. These enhancements would expand operational reach but still adhere to the same governance-first principle that distinguishes the present system.

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Acknowledgement

Success of a project like this, involving technical expertise, patience and the support of our guides, is possible because of the team's collaboration. We express our sincere gratitude to our internal guides, **Mrs. Chetana Badgujar** and **Mr. Anand Pardeshi**, for their continuous guidance and support throughout the project. We also extend our heartfelt thanks to our external mentor, **Mr. Akshay Kunkulol** (Founder & CEO, Infinity-Pool Finnotech Pvt. Ltd.), for providing the problem statement, learning material and regular feedback through the internship reviews. Finally, we thank our college, **Fr. C. Rodrigues Institute of Technology, Vashi**, for providing the opportunity and environment to learn and grow.

Project Group Members:

1. Krish Pinto, 1023228

2. Jerin Reji, 1023139

3. Ved Teredesai, 5023166

4. Samuel Rodrigues, 5023146

Appendix A: Code Samples

The following excerpts illustrate the core rubric-generation, resume-screening, persistence and approval-routing logic used by the system.

A.1 Rubric Generation from Job Description

Listing A.1: Generating and validating a structured rubric

```
export async function generateJDRubric(
  jobDescription: string
): Promise<JDRubric> {
  const trimmed = jobDescription?.trim();
  if (!trimmed || trimmed.length < 10) {
    throw new Error("Job description must be at least 10 characters long.");
  }

  try {
    const prompt = `
      Analyze the job description and return a JSON rubric containing:
      role_title, category, experience_years_required,
      must_have_skills, nice_to_have_skills, resume_rubric,
      and interview_rubric. Each rubric has 3-5 criteria whose
      integer weights sum to 100, with observable key indicators.

      JOB DESCRIPTION:
      ---
      ${trimmed}
      ---
    `;

    const generated = await generateStructuredJSON<JDRubric>(
      "You are an expert talent acquisition specialist. Return a structured rubric as JSON.",
      prompt,
      { temperature: 0.0 }
    );

    const validation = JDRubricSchema.safeParse(generated);
    if (validation.success) {
      const rubric = validation.data;
      rubric.resume_rubric = normalizeRubricWeights(rubric.resume_rubric);
      rubric.interview_rubric = normalizeRubricWeights(rubric.interview_rubric);
      return rubric;
    }
  } catch (error) {
    console.warn("Rubric generation failed; using the deterministic fallback.", error);
```

```
  }

  return generateMockJDRubric(trimmed);
}
```

A.2 Resume Screening and Score Storage

Listing A.2: Scoring a resume and persisting its result

```
const result = await generateStructuredJSON<Record<string, unknown>>(
  SYSTEM_SCREENING_PROMPT,
  buildCandidateScreeningPrompt(job, candidate, rubric, candidateName, expYears),
  { temperature: 0.1 }
);

const rubricScores =
  result.rubric_scores && typeof result.rubric_scores === "object"
    ? result.rubric_scores as Record<string, number>
    : {};

const score = calculateHybridScore(rubricScores, rubric);

const { data, error } = await supabase
  .from("applications")
  .update({
    resume_rank_score: score,
    resume_breakdown_json: { rubric_scores: rubricScores },
    recruiter_notes: String(result.overall_summary ?? ""),
  })
  .eq("job_id", jobId)
  .eq("candidate_id", candidateId)
  .neq("status", "rejected")
  .is("deleted_at", null)
  .select("id");

if (error) {
  throw new Error(`Could not save screening result: ${error.message}`);
}
if (!data?.length) {
  throw new Error("No eligible application was found to update.");
}

const { error: statusError } = await supabase
  .from("applications")
  .update({ status: "in_progress" })
  .eq("id", data[0].id)
  .eq("current_round", "Resume Screening");

```

```
if (statusError) {
  throw new Error(`Could not update application status: ${
     statusError.message}`);
}
```

A.3 Workflow Router and Approval Gate

Listing A.3: Routing HR decisions and gating interview approval

```
const body = await req.json();
const { jobId, decision, notes } = body;

if (!jobId || !["approved", "rejected", "pending"].includes(
   decision)) {
  return NextResponse.json({ success: false, error: "Invalid
     decision." }, { status: 400 });
}

const company = await getCurrentHrCompany();
if (!company) {
  return NextResponse.json({ success: false, error: "HR access
     required." }, { status: 403 });
}
if (!await ownsCompanyApplication(company.companyId, jobId,
   candidateId)) {
  return NextResponse.json({ success: false, error: "Application
     not found." }, { status: 404 });
}

if (decision === "approved") {
  const approval = await approveApplicationAndNotify({ jobId,
     candidateId, notes });
  if (!approval.transitioned && !approval.alreadyInInterview) {
    return NextResponse.json(
      { success: false, error: approval.failureReason },
      { status: 409 }
    );
  }
} else {
  await updateApplicationDecision(jobId, candidateId, decision,
     notes);
}

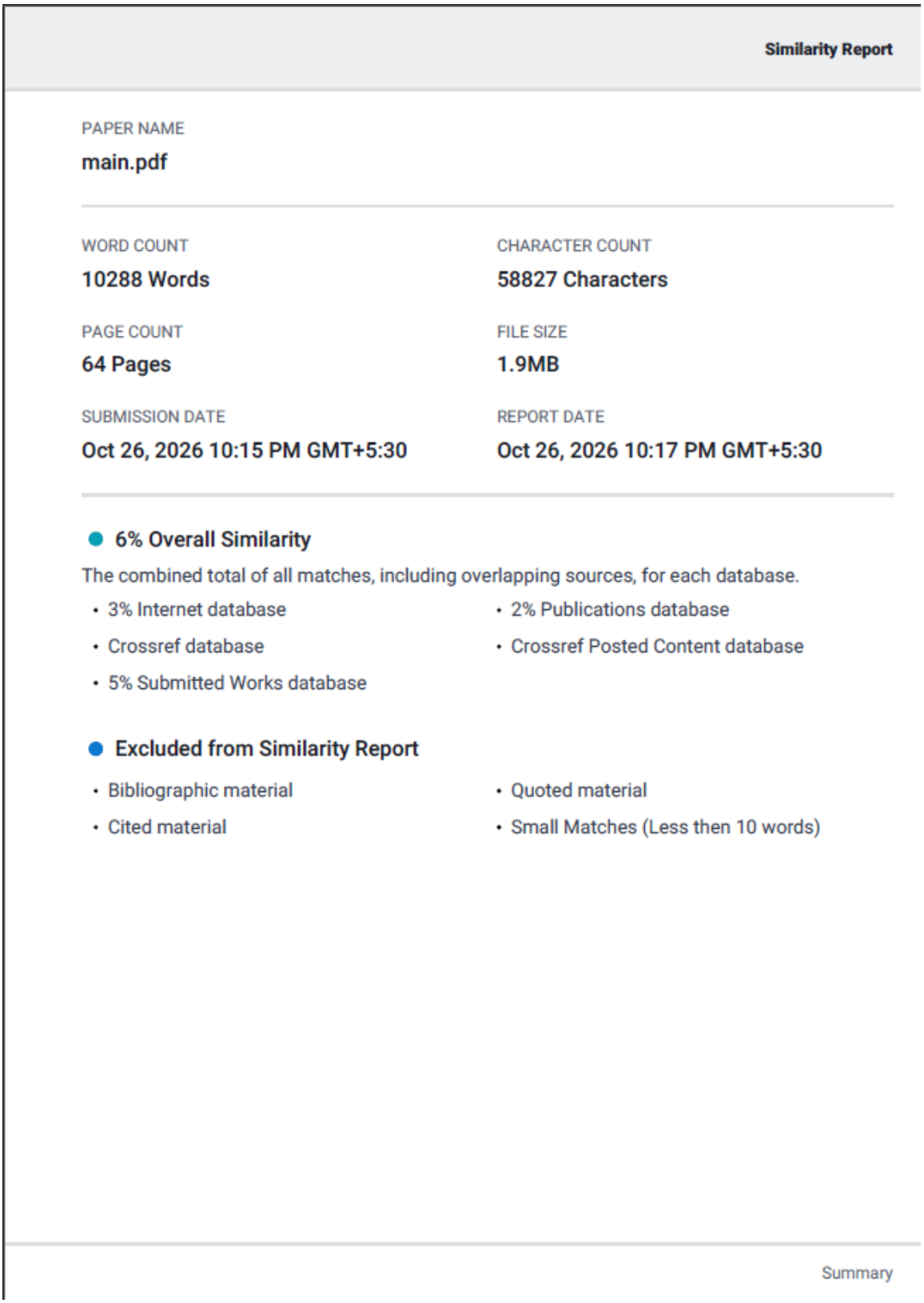
// Automatic approval is requested only after a successful
   screening score
// exceeds the job's configured threshold.
if (
  !evaluation.ai_failed &&
  Number.isFinite(evaluation.internal_score) &&
  evaluation.internal_score > job.auto_approve_threshold
) {
  await approveApplicationAndNotify({
    jobId,
```

```
    candidateId: evaluation.candidate_id,
    automatic: true,
  });
}

// The approval service claims the transition transactionally before sending
// the invitation. Automatic approvals require an open job and eligible round.
const claim = await claimApplicationForInterview({
  jobId,
  candidateId,
  notes,
  automatic,
});

if (claim.kind === "transitioned") {
  await sendCandidateApprovalEmail({ jobId, candidateId });
}
```

Appendix B: Plagiarism Report



Similarity Report

PAPER NAME

main.pdf

WORD COUNT

10288 Words

CHARACTER COUNT

58827 Characters

PAGE COUNT

64 Pages

FILE SIZE

1.9MB

SUBMISSION DATE

Oct 26, 2026 10:15 PM GMT+5:30

REPORT DATE

Oct 26, 2026 10:17 PM GMT+5:30

6% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

- 3% Internet database
- 2% Publications database
- Crossref database
- Crossref Posted Content database
- 5% Submitted Works database

Excluded from Similarity Report

- Bibliographic material
- Quoted material
- Cited material
- Small Matches (Less then 10 words)

Summary

Agnel Charities

Fr. C. Rodrigues Institute of Technology, Vashi
(An Autonomous Institute and Permanently Affiliated to University of Mumbai)

T-52

Department of Information Technology
Mini / Major Project Report Appendix C

Semester & Academic Year:	Semester VII, Academic Year 2026–27
Group Details:	Krish Pinto (1023228), Jerin Reji (1023139), Ved Teredesai (5023166), Samuel Rodrigues (5023146)
Title:	Business Function Agentic Workflows: HR, Marketing, Podcast Research & Enterprise Integrations
Key Domain / Technology:	Agentic AI and multi-agent LLM workflows; Next.js, React, TypeScript, Supabase, retrieval-augmented generation, rubric-based screening, human-in-the-loop approvals, and audit logging.
Category-I:	External – InfinityPool (Shankh) (InfinityPool Finnotech Pvt Ltd).
Category-II:	Application / Product – a prototype platform for supervised business workflows across HR, marketing, podcast research, and enterprise integrations.
SDG:	SDG 1, SDG 2, SDG 4
Mapping to POs:	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11
Mapping to PSOs:	PSO1, PSO2, PSO3
Remarks:	

Name and Signature of the Guide

Mr. Anand Pardeshi

Note: The guide should write remarks based on the observation of the project's objectives, achievements, and outcomes.

Appendix D: Internship Commencement Letters



InfinityPool Finnotech Private Limited

CIN U74999PN2022PTC213334
Reg. Address: 603, Solitaire Five, Bibwewadi, Pune-37, MH, India
Corp. Address: World Trade Center, Tower 5, Kharadi, Pune-14, MH, India
Web: InfinityPool.in Phone: +91 980 1008 108

Date: 16 June 2026

INTERNSHIP COMMENCEMENT LETTER

To,
Samuel Rodrigues,
FCRIT, Vashi, Navi Mumbai

Dear Samuel Rodrigues,

In accordance with the Memorandum of Understanding (MoU) between InfinityPool Finnotech Pvt Ltd (IFPL) and Agnel Charities' Fr. Conceicao Rodrigues Institute of Technology (FCRIT), we would like to extend our heartiest congratulations on your selection as "Software Engineering Intern" with IFPL. Your internship shall commence on June 16, 2026 and shall end on May 30, 2027 (the "Term"). You will be required to report to Mr. Mahadev Walekar during this internship. Our team at IFPL looks forward to having you work with us.

During your internship, you will be working on applying Artificial Intelligence techniques to solve problems in the Finance domain. Complete details of the project have been separately shared with you by IFPL. Your participation and contributions to the project will be subject to the terms and conditions of the non-disclosure agreement signed by you with IFPL. Any breach found in this regard will result in permissible action and immediate termination of the project. You may work from home during this Term. On successful completion of the internship, your supervisor may recommend you for a full-time offer of employment.

Once again, congratulations on your selection, and all the best for your endeavors.

Sincerely,



Akshay Kunkulol
Founder & CEO

For verification of this document, please email contact@infinitypool.in

InfinityPool Finnotech Private Limited
CIN U74999PN2022PTC213334
Reg. Address: 603, Solitaire Five, Bibwewadi, Pune-37, MH, India
Corp. Address: World Trade Center, Tower 5, Kharadi, Pune-14, MH, India
Web: InfinityPool.in Phone: +91 980 1008 108

Date: 16 June 2026

INTERNSHIP COMMENCEMENT LETTER

To,
Ved Teredesai,
FCRIT, Vashi, Navi Mumbai

Dear Ved Teredesai,

In accordance with the Memorandum of Understanding (MoU) between InfinityPool Finnotech Pvt Ltd (IFPL) and Agnel Charities' Fr. Conceicao Rodrigues Institute of Technology (FCRIT), we would like to extend our heartiest congratulations on your selection as "Software Engineering Intern" with IFPL. Your internship shall commence on June 16, 2026 and shall end on May 30, 2027 (the "Term"). You will be required to report to Mr. Mahadev Walekar during this internship. Our team at IFPL looks forward to having you work with us.

During your internship, you will be working on applying Artificial Intelligence techniques to solve problems in the Finance domain. Complete details of the project have been separately shared with you by IFPL. Your participation and contributions to the project will be subject to the terms and conditions of the non-disclosure agreement signed by you with IFPL. Any breach found in this regard will result in permissible action and immediate termination of the project. You may work from home during this Term. On successful completion of the internship, your supervisor may recommend you for a full-time offer of employment.

Once again, congratulations on your selection, and all the best for your endeavors.

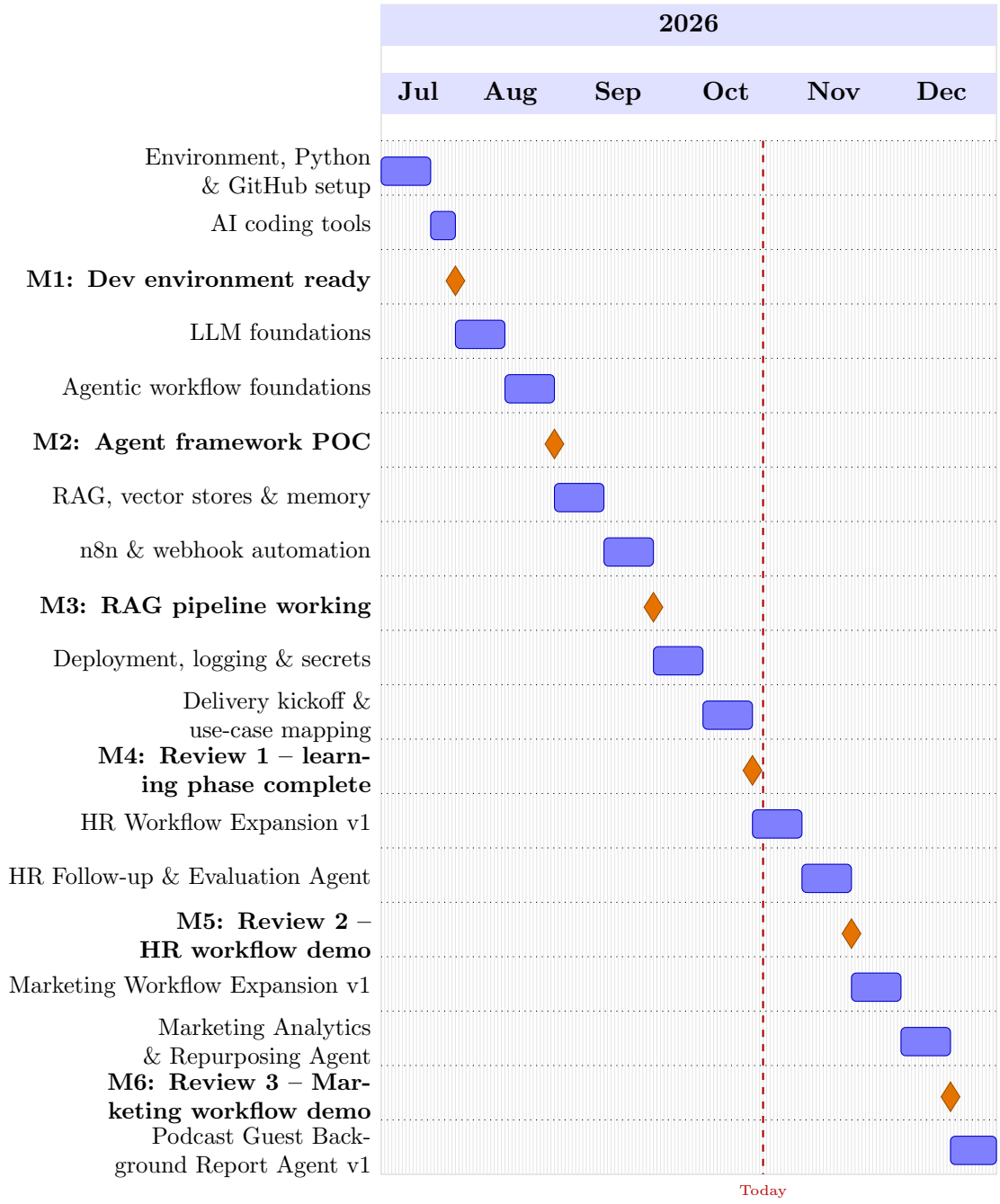
Sincerely,



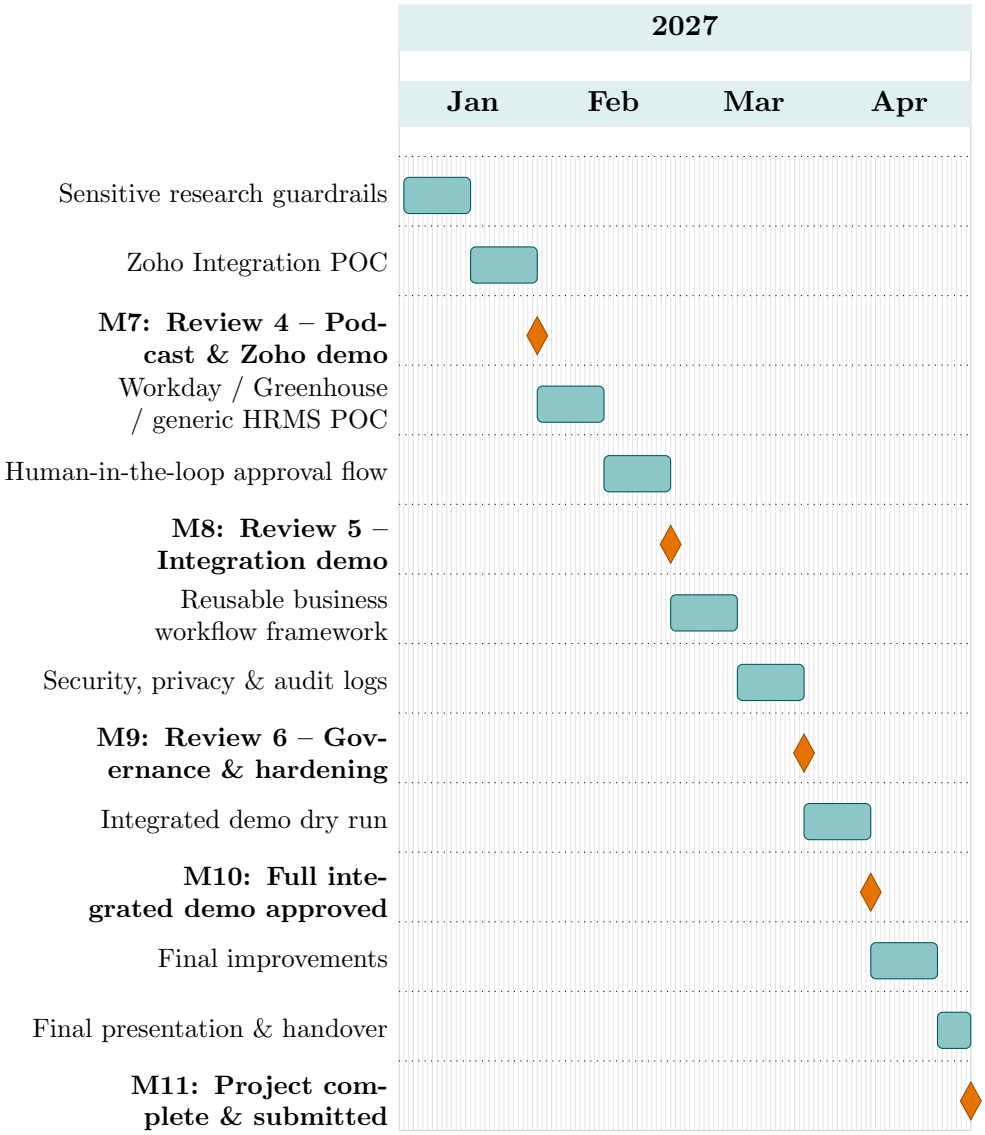
Akshay Kunkulol
Founder & CEO

For verification of this document, please email contact@infinitypool.in

Appendix E: Timeline Chart



Timeline Chart of the Project Development – Semester 7 (July–December 2026)



Timeline Chart of the Project Development – Semester 8 (January–April 2027)

Appendix F: Software Requirements Specification

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for **Business Function Agentic Workflows**, an integrated multi-agent AI platform for HR recruitment, marketing content operations, podcast guest research and enterprise integrations. The document is intended for the development team, academic supervisors and industry partners at InfinityPool (Shankh).

1.2 Scope

Business Function Agentic Workflows automates repetitive business workflows across HR, marketing and podcast production using specialised LLM agents coordinated by a workflow router. The system provides rubric-based resume screening, AI-conducted interviews, channel-specific content drafting, source-backed guest briefings and secure connectors to Zoho, Workday-style and Greenhouse-style APIs. It does not make final hiring, rejection, salary or publishing decisions autonomously; every consequential action requires human approval.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
ATS	Applicant Tracking System
HRMS	Human Resource Management System
PII	Personally Identifiable Information
OAuth	Open Authorisation (industry-standard protocol for access delegation)
JD	Job Description
API	Application Programming Interface
HR	Human Resources
CRM	Customer Relationship Management
n8n	Node-based workflow automation platform
pgvector	PostgreSQL extension for vector similarity search

1.4 References

Industry partner milestone plan and problem statement provided by InfinityPool (Shankh); fortnightly review presentations (July–October 2026); IEEE Std 29148-2018 Systems and Software Engineering – Life cycle processes – Requirements engineering.

2. Overall Description

2.1 Product Perspective

Business Function Agentic Workflows is a standalone web-based platform that integrates with external enterprise systems through a connector layer. The front end is a Next.js dashboard; the back end comprises a workflow router, domain-specific LLM agents, a RAG knowledge layer, a compliance reviewer, a human approval gate and an audit store backed by Supabase. The system does not replace existing HRMS or ATS products; it orchestrates work across them.

2.2 Product Functions

The major functions of the system are:

- Generation of rubrics from a job description and their subsequent approval by HR.

- Resume parsing, adversarial-content scanning, claim verification, and rubric-based scoring.
- AI-conducted live interviews with rubric-aligned scoring and final candidate ranking.
- HR assistant to provide support on onboarding, leave, attendance and policy-related queries.
- Content calendar, campaign brief and channel-specific draft creation for marketing teams.
- Podcast guest background reports, which have all claims marked out as fact, opinion or allegation.
- Synchronisation of enterprise data with Zoho People, Zoho CRM and Recruit, Workday-style APIs and Greenhouse-style ATS objects.
- Human approval queues, redaction of PII, compliance review and immutable audit logging.

2.3 User Characteristics

- **HR Recruiter:** Intermediate computer literacy; creates jobs, reviews rubrics and ranking lists, makes hiring decisions.
- **Candidate:** General public; submits a resume and completes an AI interview via a public link.
- **Marketing User:** Non-technical; creates content themes and manages the approval queue.
- **Podcast Host / Researcher:** Moderate technical literacy; requests and reviews guest briefings.
- **Administrator:** Technical; configures connectors, OAuth accounts and system settings.

2.4 Constraints

- Development uses only synthetic, publicly available or sanitised data; no live client PII.
- Enterprise integrations use sandbox, trial or mock APIs during development.

- Hosted LLM costs must be minimised; local models via Ollama are used for development and testing.
- The system must run on the developers' existing hardware for development; production deployment targets free or low-cost hosting tiers.

2.5 Assumptions and Dependencies

- Google Gemini and OpenAI API endpoints are available and responsive.
- Supabase PostgreSQL, Storage and pgvector are accessible with the configured credentials.
- Zoho, Workday-style and Greenhouse-style sandbox APIs maintain their documented schema.
- n8n community edition is self-hosted and reachable from the backend.

3. Specific Requirements

3.1 External Interfaces

User Interfaces: A Next.js web dashboard with separate views for HR recruitment, marketing, podcast research, integrations, human review and audit log. A public candidate apply page accessible without login. Approval queue items show the full agent output, evidence and approve/edit/reject actions.

Software Interfaces: Supabase REST and Realtime APIs for data, storage, authentication and vector search; Google Gemini and OpenAI completion and embedding endpoints; Zoho People and Recruit APIs; Workday-style HRIS API stub; Greenhouse Harvest API; Meta Marketing API; Mailchimp Marketing API; Google Calendar and Sheets APIs; n8n webhook endpoints.

Hardware Interfaces: No direct hardware interfaces; the system is entirely web-based.

Communication Interfaces: HTTPS for all client–server and server–API communication; OAuth 2.0 for enterprise connector authentication; webhook payloads over HTTPS for event-driven flows.

3.2 Functional Requirements

ID	Requirement
REQ-01	The system shall classify every incoming request and route it to the correct HR, marketing, podcast research or integration workflow.
REQ-02	The system shall generate resume and interview rubrics from a job description and present them for HR review and approval before use.
REQ-03	The system shall parse uploaded resumes, scan them for adversarial content, verify material claims against public sources and score them against the approved rubric with evidence citations.
REQ-04	Candidates whose resume score exceeds the HR-configured threshold shall be auto-qualified; others shall be queued for HR approval.
REQ-05	The system shall conduct live AI interviews, score each answer against the interview rubric and produce a ranked shortlist with scorecards for HR review.
REQ-06	The system shall answer employee queries about onboarding, leave, attendance and HR policy by retrieving from company documents and Zoho People data, citing the source.
REQ-07	The system shall generate marketing content calendars, campaign briefs, audience segments and channel-specific post drafts from a provided theme or source content, guided by a stored brand knowledge base.
REQ-08	All marketing drafts shall be held in an approval queue; the system shall publish only after a human approves.
REQ-09	The system shall produce podcast guest briefings covering career timeline, expertise map, prior interviews, suggested questions and sensitive topics, with every claim attributed to a cited source and labelled as a verified fact, allegation or opinion.
REQ-10	The system shall read and write data in Zoho, Workday-style and Greenhouse-style systems through a typed connector interface that handles OAuth, pagination, rate limits, retries and idempotency.
REQ-11	All external write operations and public communications, except auto-qualified interview invitations, shall wait for explicit human approval recorded in the audit log.

3.3 Usability Requirements

- HR and marketing staff with no AI background shall be able to use the dashboard and approval queues without training beyond a brief orientation.
- Approval queue items must display the full agent rationale and evidence so that a reviewer can make an informed decision without consulting other systems.
- The candidate apply page and interview interface must be accessible to applicants on standard desktop and mobile browsers without any account creation.

3.4 Performance Requirements

- Resume screening results shall be produced within 60 seconds of submission for a standard two-page resume.
- AI interview responses shall be generated within 5 seconds of each candidate answer to maintain a conversational pace.
- Marketing draft generation for a five-post calendar shall complete within 90 seconds.
- Connector sync operations shall complete within 30 seconds for standard paginated fetches of up to 100 records.

3.5 Logical Data Requirements

Jobs, rubrics, candidates, applications, scores, interview transcripts, marketing campaigns, post drafts, guest briefings, connector accounts and audit events are stored in Supabase PostgreSQL. Resume files and multimedia attachments are stored in Supabase private storage buckets. Embeddings for the knowledge base are indexed using pgvector. Retention of candidate PII is limited to the duration of the hiring process; deletion requests are honoured within the same working day.

3.6 Security and Safety Requirements

- API keys, OAuth tokens and database credentials must be stored in environment variables and must never appear in source code, prompts or logs.
- Candidate PII must be redacted from LLM prompts before scoring; only anonymised content is sent to external model APIs.

- Resume files must be stored in private buckets accessible only through signed, time-limited URLs.
- All connector write operations require an HR-approved scope; least-privilege OAuth scopes must be used.
- Resumes must be scanned for hidden instructions before scoring; flagged files must be routed to a human review queue.
- Role-based access control must prevent candidates from accessing recruiter views and prevent marketing users from accessing HR data.

3.7 Quality Attributes / Non-Functional Requirements

- **Reliability:** Connector calls must implement exponential back-off retries and idempotency keys so that transient failures and duplicate events do not create duplicate records.
- **Maintainability:** New agents, connectors and business functions must be addable through the reusable workflow framework without modifying the router core.
- **Evaluability:** Each workflow must be exercisable against a synthetic test suite; accuracy, hallucination rate, latency and task completion rate must be measurable.
- **Portability:** The backend must be deployable on any Node.js-compatible hosting platform; connector configuration must be isolated in environment variables.
- **Auditability:** Every output must be traceable from the final decision back to the initiating prompt, tool calls, retrieved sources and reviewer within the audit log.

4. Verification / Acceptance Criteria

Req. ID	Method	Criterion
REQ-01	Test execution	Given ten sample requests across all four domains, the router correctly classifies all ten.
REQ-02	Demonstration	HR user uploads a JD and receives a structured rubric within 30 seconds; rubric is editable and saveable.
REQ-03	Test execution	On a synthetic resume containing a hidden instruction, the scanner flags the resume before scoring.
REQ-04	Test execution	A resume scoring above the threshold is auto-qualified; one below requires HR action.
REQ-05	Demonstration	An end-to-end AI interview for a test candidate produces a scorecard with per-criterion evidence.
REQ-06	Test execution	On five synthetic employee queries, the HR assistant returns answers citing the correct policy clause or Zoho record.
REQ-07	Demonstration	Given a theme, the system produces a five-post calendar draft within 90 seconds.
REQ-08	Test execution	No post is published when the approval queue item is in a pending state.
REQ-09	Demonstration	A guest briefing is produced with claims labelled as fact, allegation or opinion, each with a cited URL.
REQ-10	Test execution	A connector read against the Zoho sandbox returns paginated results without duplicate entries on retry.
REQ-11	Inspection	Code review confirms no write connector call is made without a prior approval record in the audit log.
REQ-12	Inspection	Audit log entries for a sample run contain prompt hash, tool calls, source URLs, reviewer ID and final output ID.

5. Supporting Information

Appendix A – Traceability Matrix

Stakeholder Need	Mapped Requirements
Reduce recruiter time on resume review	REQ-02, REQ-03, REQ-04, REQ-12
Fair, explainable candidate scoring	REQ-02, REQ-03, REQ-05, REQ-12
Prevent adversarial resume manipulation	REQ-03, REQ-06 (security scan)
Maintain brand consistency in marketing	REQ-07, REQ-08
Fact-safe podcast guest briefings	REQ-09
Secure enterprise data exchange	REQ-10, REQ-11, REQ-06 (security)
Full audit trail for every AI decision	REQ-11, REQ-12

Appendix B – Change History

Version	Date	Change Summary
1.0	Oct 2026	Initial SRS drafted from problem statement, milestone plan and existing-systems study.